

Distributed Target Tracking under Partial Feedback using Lyapunov-based Deep Neural Networks

Cristian F. Nino, Omkar Sudhir Patil, Sage C. Edwards, Zachary I. Bell and Warren E. Dixon

Abstract—The target tracking problem is addressed for multi-agent systems where the target state information is only partially available to the agents via a heterogeneous measurement model. A necessary and sufficient condition, termed trackability, is provided, to indicate the feasibility for tracking a target with partial measurements. A Lyapunov-based deep neural network (Lb-DNN) adaptive controller is developed to achieve target tracking, under the trackability condition, by adaptively compensating for the uncertainty stemming from the unknown target dynamics. A Lyapunov-based stability analysis is provided to guarantee exponential target state estimation and tracking within a neighborhood of the target state.

Index Terms—Multi-agent systems, Adaptive Control, Neural Networks

I. INTRODUCTION

INTEREST in the control domain due to their extensive applications (e.g., consensus, wireless sensor networks, formation control, and cooperative robotics [1]–[4]). An important application of multi-agent systems is the use of collaborative agents to track an object of interest.

Target tracking may involve multiple agents attempting to monitor, engage, or intercept a target, which could be semi-cooperative or adversarial, as seen in search-and-rescue operations [5] and direct action objectives [6], respectively. However, many studies in this area presuppose knowledge of the target state. In spatial tasks, a challenge arises from limited overall knowledge about the target or reduced state accuracy due to measurement errors, necessitating a method for state reconstruction from partially available state information to enable effective tracking (e.g., [7], [8]). Moreover, in adversarial target tracking scenarios, the target may actively try to evade and be uncooperative with respect to the controlled agents. In such cases, simply estimating the target's state may not be sufficient. Learning the target's dynamics enables the prediction of its future behavior, allowing for more effective tracking strategies such as interception.

Cristian F. Nino, Omkar Sudhir Patil, Sage C. Edwards, and Warren E. Dixon are with the Department of Mechanical and Aerospace Engineering, University of Florida, Gainesville, FL, 32611, USA. Email: {cristian1928, patilomkarsudhir, sageedwards, wdixon}@ufl.edu.

Zachary I. Bell is with the Munitions Directorate, Air Force Research Laboratory, Eglin AFB, FL 32542, USA. Email: zachary.bell.10@us.af.mil.

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Adaptive control methods present potential solutions to these challenges. Recently, there has been an increasing interest in adaptive control for nonlinear-in-parameter (NIP) uncertainty systems. Deep Neural Networks (DNNs) exhibit a universal function approximation property (e.g., [9]) and have shown significant empirical and experimental improvements over traditional NN methods (e.g., [10], [11]). Traditional limitations of DNNs that resulted from offline training through numerical optimization methods (e.g., [12]) have recently been addressed through Lyapunov-based (Lb-) DNNs. Lb-DNNs (e.g., [13]–[17]) have analytical update laws derived from the Lyapunov-based stability analysis, which yield convergence/boundedness guarantees. Because the adaptation laws are analytic, they execute in real-time with feedback, yielding lifelong learning without the requirement for pretraining.

While results such as [18]–[21] have employed shallow neural networks (NNs) for the distributed tracking problem, few results have concurrently addressed heterogeneous sensing, state estimation, and function approximation. Specifically, [22] is the inaugural distributed state estimation result for uncertain nonlinear systems using DNNs. The work in [22], focused on wireless sensor networks, pioneered the use of a DNN architecture that updated its outer layer weights using a Lyapunov-based update law, while iteratively updating the inner layer weights using discrete batch updates over finite time intervals. Furthermore, a distributed observer was developed to estimate the unknown state of a target system, with each agent in the network potentially having different sensing capabilities. Despite its significant contribution, this work did not concentrate on the target tracking problem and did not utilize real-time adaptation for all layers of the DNN (e.g., [14]), addressing only the target state estimation problem. There are additional mathematical challenges that arise in the Lyapunov-based design while achieving the combined objectives of distributed state estimation and target tracking.

The approach developed in this paper presents the first result on target tracking for multi-agent systems where the target state information is only partially available to the agents from a heterogeneous measurement model. A condition termed trackability is presented to determine whether it is possible to track a target using only partial measurements. To learn the dynamics of the target, an Lb-DNN is developed, where the weights of all layers of the DNN are updated in real-time. To facilitate the tracking objective, a distributed

observer is designed which employs an Lb-DNN to learn the state of the target. The Lb-DNN state estimate is combined with a distributed controller to achieve the target tracking objective. A Lyapunov-based stability analysis is used to prove exponential convergence of the target state estimates and agent states to a neighborhood of the target state, provided the trackability condition is satisfied.

II. NOTATION AND PRELIMINARIES

The $p \times p$ identity matrix and the $p \times 1$ column vector of ones are denoted by I_p and $\mathbf{1}_p$, respectively. Given $M \in \mathbb{Z}_{>0}$, the enumeration operation $[\cdot]$ is defined as $[M] \triangleq \{1, 2, \dots, M\}$. The Kronecker product of $A \in \mathbb{R}^{p \times q}$ and $B \in \mathbb{R}^{m \times n}$ is denoted by $A \otimes B \in \mathbb{R}^{pm \times qn}$. The right-to-left matrix product operator is represented by $\prod_{p=1}^m$, i.e., $\prod_{p=1}^m A_p = A_m \dots A_2 A_1$ and $\prod_{p=a}^m A_p = I$ if $a > m$. The kernel of a matrix $A \in \mathbb{R}^{m \times n}$, denoted by $\ker(A)$, is the set of all column vectors $\mathbf{x}$ of size $n \times 1$ such that $A\mathbf{x} = \mathbf{0}$, where $\mathbf{0}_m$ is the zero vector of size $m \times 1$. The maximum and minimum eigenvalues of $G = G^T$ are denoted by $\lambda_{\max}(G) \in \mathbb{R}$ and $\lambda_{\min}(G) \in \mathbb{R}$, respectively. Given a positive integer N and collection $\{x_i\}_{i \in [N]} \subset \mathbb{R}^n$, let $(x_i)_{i \in [N]} \triangleq [x_1^T, x_2^T, \dots, x_N^T]^T \in \mathbb{R}^{nN}$. Given $H \in \mathbb{R}^{m \times m}$ with columns $\{h_i\}_{i \in [n]} \subset \mathbb{R}^m$, $\text{vec}(H) = [h_1^T, h_2^T, \dots, h_n^T]^T \in \mathbb{R}^{mn}$. The Euclidean norm of $r \in \mathbb{R}^n$ is $\|r\| \triangleq \sqrt{r^T r}$. The Frobenius norm is denoted by $\|\cdot\|_F \triangleq \|\text{vec}(\cdot)\|$. Given any $A \in \mathbb{R}^{p \times a}$, $B \in \mathbb{R}^{a \times r}$, and $C \in \mathbb{R}^{r \times s}$, the vectorization operator satisfies the property $\text{vec}(ABC) = (C^T \otimes A)\text{vec}(B)$. Differentiating on both sides with respect to $\text{vec}(B)$ yields the property

$$\frac{\partial}{\partial \text{vec}(B)} \text{vec}(ABC) = (C^T \otimes A). \quad (1)$$

The space of essentially bounded Lebesgue measurable functions is denoted by $\mathcal{L}_\infty$. Given $w \in \mathbb{R}$ and some functions f and g , the notation $f(w) = \mathcal{O}^m(g(w))$ means that there exists some constants $M \in \mathbb{R}_{>0}$ and $w_0 \in \mathbb{R}$ such that $\|f(w)\| \leq M \|g(w)\|^m$ for all $w \geq w_0$.

A. Graphs

Let $\mathcal{G} \triangleq (\mathcal{V}, E)$ represent a static and undirected graph with number of nodes $N \in \mathbb{Z}_{\geq 2}$, where the node set is denoted by $\mathcal{V} \triangleq [N]$, and the edge set is denoted by $E \subseteq \mathcal{V} \times \mathcal{V}$. An edge between nodes i and k belongs to the edge set (i.e., $(i, k) \in E$) if and only if node i can send information to node k . Since the graph $\mathcal{G}$ is undirected, $(i, k) \in E$ if and only if $(k, i) \in E$. An undirected graph is connected whenever there exists a sequence of edges in E linking any two distinct nodes. The neighborhood set of node i is $\mathcal{N}_i \triangleq \{k \in \mathcal{V} \setminus \{i\} : (k, i) \in E\}$. Let $\mathcal{A} \triangleq [a_{ik}] \in \mathbb{R}^{N \times N}$ be the adjacency matrix of $\mathcal{G}$, where $a_{ik} = 1$ if $(k, i) \in E$ and $a_{ik} = 0$ otherwise. Within this work, no self-loops are considered. Therefore, $a_{ii} \triangleq 0$ for all $i \in \mathcal{V}$. The degree matrix of $\mathcal{G}$ is $\mathcal{D} \triangleq \text{diag}(\mathcal{A} \cdot \mathbf{1}_N) \in \mathbb{R}^{N \times N}$. Using the degree and adjacency matrices, the Laplacian matrix of the graph $\mathcal{G}$ is $\mathcal{L}_\mathcal{G} \triangleq \mathcal{D} - \mathcal{A}$.

B. Deep Neural Network Model

Let $\kappa \in \mathbb{R}^{L_0}$ denote the DNN input with size $L_0 \in \mathbb{Z}_{>0}$, and $\theta \in \mathbb{R}^p$ denote the vector of DNN parameters (i.e., weights and bias terms) with size $p \in \mathbb{Z}_{>0}$. Then, a fully-connected feed-forward DNN $\Phi(\kappa, \theta)$ with $k \in \mathbb{Z}_{>0}$ hidden layers and output size $L_{k+1} \in \mathbb{Z}_{>0}$ is defined using a recursive relation $\varphi_j \in \mathbb{R}^{L_{j+1}}$ modeled as

$$\varphi_j \triangleq \begin{cases} V_{j+1}^\top \kappa_a, & j = 0, \\ V_{j+1}^\top \phi_j(\varphi_{j-1}) & j \in \{1, \dots, k\}, \end{cases} \quad (2)$$

where $\Phi(\kappa, \theta) = \varphi_k$, $\kappa_a \triangleq [\kappa^\top, 1]^\top$ denotes the augmented input that accounts for the bias terms, $L_j \in \mathbb{Z}_{>0}$ denotes the number of neurons in the j^{th} layer with $L_j^a \triangleq L_j + 1$, and $V_{j+1} \in \mathbb{R}^{L_j^a \times L_{j+1}}$ denotes the matrix of weights and biases, for all $j \in \{0, \dots, k\}$.

The vector of activation functions is denoted by $\phi_j : \mathbb{R}^{L_j} \rightarrow \mathbb{R}^{L_j^a}$ for all $j \in \{1, \dots, k\}$. The vector of activation functions can be composed of various activation functions and hence may be represented as $\phi_j = [\varsigma_1, \dots, \varsigma_{L_j}, 1]^\top$ for all $j \in \{1, \dots, k\}$, where $\varsigma_j : \mathbb{R} \rightarrow \mathbb{R}$ for all $j \in \{1, \dots, L_j\}$ denotes a piecewise continuously differentiable activation function and the 1 denotes the augmented hidden layer that accounts for the bias term. For the DNN architecture in (2), the vector of DNN weights is $\theta \triangleq [\text{vec}(V_1)^\top, \dots, \text{vec}(V_k)^\top]^\top$ with size $p = \sum_{j=0}^k L_j^a L_{j+1}$.

Consider $y_j \in \mathbb{R}^{L_j}$ where $y_j = [y_1, \dots, y_{L_j}]$ with $y_i \in \mathbb{R}$ for all $i \in \{1, \dots, L_j\}$. The Jacobian $\frac{\partial \phi_j}{\partial y_j} : \mathbb{R}^{L_j} \rightarrow \mathbb{R}^{L_j^a \times L_j}$ of the activation function vector at the j^{th} layer is given by $[\varsigma'_1(y_1)\mathbf{e}_1, \dots, \varsigma'_{L_j}(y_{L_j})\mathbf{e}_{L_j}, \mathbf{0}_{L_j}]^\top \in \mathbb{R}^{L_j^a \times L_j}$, where ς'_j denotes the derivative of ς_j with respect to its argument for $j \in \{1, \dots, L_j\}$, $\mathbf{e}_i$ is the i^{th} standard basis vector in $\mathbb{R}^{L_j}$, and $\mathbf{0}_{L_j}$ is the zero vector in $\mathbb{R}^{L_j}$.

Let the gradient of the DNN with respect to the weights be denoted by $\nabla_\theta \Phi(\kappa, \theta) \triangleq \frac{\partial}{\partial \theta} \Phi(\kappa, \theta)$, which can be represented as $\nabla_\theta \Phi(\kappa, \theta) = \left[\frac{\partial}{\partial \text{vec}(V_1)} \Phi(\kappa, \theta), \dots, \frac{\partial}{\partial \text{vec}(V_k)} \Phi(\kappa, \theta) \right] \in \mathbb{R}^{L_{k+1} \times p}$, where $\frac{\partial}{\partial \text{vec}(V_j)} \Phi(\kappa, \theta) \in \mathbb{R}^{L_{k+1} \times L_j^a L_j}$ for all $j \in \{1, \dots, k+1\}$. Then, using (2) and the property of the vectorization operator in (1) yields

$$\nabla_\theta \Phi(\kappa, \theta) = \left(\prod_{\ell=j+1}^k V_{\ell+1}^\top \frac{\partial \phi_\ell}{\partial \varphi_{\ell-1}} \right) (I_{L_{j+1}} \otimes \varrho_j), \quad (3)$$

for $j \in \{0, \dots, k\}$, where $\varrho_j = \kappa_a^\top$ if $j = 0$ and $\varrho_j = \phi_j^\top(\varphi_{j-1})$ if $j \in \{1, \dots, k\}$.

Theorem 1. (Universal Approximation Theorem [9]) Let $\phi : \mathbb{R} \rightarrow \mathbb{R}$ be any smooth function and let Φ denote the space of feed-forward neural networks with fixed width and arbitrary depth, such that every hidden neuron has activation function ϕ and every output neuron has the identity as its activation function. Fix $f \in C(\mathbb{R}^n, \mathbb{R}^m)$, $\bar{\epsilon} > 0$, and fix a compact $\Omega \subseteq \mathbb{R}^n$. There exists $\Phi(\kappa, \theta) \in \Phi$ such that $\sup_{x \in \Omega} \|\Phi(\kappa, \theta) - f(x)\| < \bar{\epsilon}$.

III. PROBLEM FORMULATION

A. System Dynamics and Network Topology

Consider a target system with dynamics governed by the model

$$\dot{q}_0 = f(q_0), \quad (4)$$

where $q_0, \dot{q}_0 \in \mathbb{R}^n$ denote the system's unknown generalized position and velocity, respectively, and the function $f : \mathbb{R}^n \rightarrow \mathbb{R}^n$ is an unknown locally Lipschitz function. Consider a network of N agents, indexed by $\mathcal{V}$, modeled by the connected and undirected communication graph $\mathcal{G} \triangleq (\mathcal{V}, E)$. The model for agent $i \in \mathcal{V}$ is given by

$$\dot{q}_i = u_i, \quad y_i = C_i(q_0 - q_i), \quad (5)$$

where $q_i \in \mathbb{R}^n$ denotes the position, $u_i \in \mathbb{R}^n$ denotes a control input, $y_i \in \mathbb{R}^{m_i}$ denotes the measured output, and $C_i \in \mathbb{R}^{m_i \times n}$ denotes the known output matrix, representing the agents' heterogeneous sensing capabilities.

Assumption 1. The position of the target agent is bounded in the sense that there exists a known constant $\bar{q}_0 \in \mathbb{R}_{>0}$ such that $\|q_0(t)\| \leq \bar{q}_0$, for all $t \in [t_0, \infty)$, where $t_0 \in \mathbb{R}_{>0}$.

Assumption 2. The unknown drift f in (4) is locally Lipschitz with a known Lipschitz constant $L \in \mathbb{R}_{>0}$, defined on $\mathcal{E} \triangleq \{\zeta \in \mathbb{R}^n : \|\zeta\| \leq \bar{q}_0 + \chi\}$, where $\chi \in \mathbb{R}_{>0}$ is a known bounding constant.

B. Objective

The primary objective is to develop a set of distributed controllers that regulate each agent towards the target, relying solely on the relative linear measurement model in (5). Given the agents' limited ability to measure the full state of the target, a secondary objective is to design an observer capable of reconstructing the unknown nonlinear system in (4) via output feedback. Given the unknown model in (4), the distributed observer will employ online learning techniques to estimate f .

To quantify the target regulation objective, let the tracking error be defined as

$$e_i \triangleq q_0 - q_i, \quad (6)$$

for all $i \in \mathcal{V}$. Given that the tracking error in (6) includes the target's unknown state, two additional errors are introduced to aid in the reconstruction of the known tracking error. To address the state reconstruction objective, the unknown state estimation error is defined as

$$\tilde{q}_i \triangleq q_0 - \hat{q}_0^i, \quad (7)$$

for all $i \in \mathcal{V}$, where $\hat{q}_0^i \in \mathbb{R}^n$ is agent i 's estimated state of q_0 . To quantify regulation of each agent towards the state estimate, the state estimation regulation error is defined as

$$\hat{e}_i \triangleq \hat{q}_0^i - q_i, \quad (8)$$

for all $i \in \mathcal{V}$. Using (6)-(8), e_i can be expressed as

$$e_i = \hat{e}_i + \tilde{q}_i. \quad (9)$$

Since the relative linear measurement model in (5) results in only partial target state measurements, a transformation is constructed to remove dependency on the partial target state. Let the output estimation error be defined as

$$\tilde{y}_i = y_i - \hat{y}_i, \quad (10)$$

where $\hat{y}_i \triangleq C_i \hat{q}_0^i$, for all $i \in \mathcal{V}$. This error is known and reflects the observer's performance based on the definition of $\hat{y}_i$.

By substituting the output from (5) into (10) and incorporating (7), it follows that

$$\tilde{y}_i = C_i \tilde{q}_i - C_i q_i. \quad (11)$$

Adding $C_i q_i$ to both sides of (11), multiplying the resulting equation by $C_i^\top$ and defining the matrix $\mathcal{C}_i \triangleq C_i^\top C_i \in \mathbb{R}^{n \times n}$ yields the computable state estimation error

$$C_i \tilde{q}_i = C_i^\top \tilde{y}_i + \mathcal{C}_i q_i. \quad (12)$$

IV. CONTROL SYNTHESIS

To facilitate the distributed control and observer design, substituting (4) into the time-derivative of (7) yields

$$\dot{\tilde{q}}_i = f(q_0) - \dot{\hat{q}}_0^i. \quad (13)$$

Since the goal is to apply an adaptive DNN-based learning technique for estimating the function f , it is essential to be able to reconstruct this unknown function using only known signals. However, since the state of the target, q_0 , is unknown, directly learning $f(q_0)$ is not feasible. To address this challenge, a reconstructable signal is introduced by adding and subtracting $f(\hat{q}_0^i)$ to (13) and defining $\tilde{f}_i \triangleq f(q_0) - f(\hat{q}_0^i)$, yielding

$$\dot{\tilde{q}}_i = f(\hat{q}_0^i) - \dot{\hat{q}}_0^i + \tilde{f}_i. \quad (14)$$

Since the function f in (14) represents an unstructured uncertainty, each agent employs an Lb-DNN to adaptively compensate for $f(\hat{q}_0^i)$ as

$$f(\hat{q}_0^i) = \Phi_i(\hat{q}_0^i, \theta_i^*) + \epsilon_i(\hat{q}_0^i), \quad (15)$$

where $\epsilon_i : \mathbb{R}^n \rightarrow \mathbb{R}^n$ is an unknown function representing the reconstruction error that can be bounded as $\sup_{\hat{q}_0^i \in \Omega} \|\epsilon_i(\hat{q}_0^i)\| \leq \bar{\epsilon}_i$, $\theta_i^* \in \mathbb{R}^p$, and $\Phi_i : \mathbb{R}^n \times \mathbb{R}^p \rightarrow \mathbb{R}^n$.

The distributed controller and observer are designed as

$$u_i = \dot{\hat{q}}_0^i + k_2 \left(C_i \hat{e}_i - \sum_{j \in \mathcal{N}_i} (\hat{e}_j - \hat{e}_i) \right), \quad (16)$$

and

$$\dot{\hat{q}}_0^i = \Phi_i(\hat{q}_0^i, \hat{\theta}_i) + k_1 \left(\sum_{j \in \mathcal{N}_i} (\hat{q}_0^j - \hat{q}_0^i) + C_i^\top \tilde{y}_i + C_i q_i \right), \quad (17)$$

respectively, where $k_1, k_2 \in \mathbb{R}_{>0}$ are user-defined gains and $\hat{\theta}_i \in \mathbb{R}^p$, for all $i \in \mathcal{V}$. Substituting (5) and (16) into the time-derivative of (8) yields

$$\dot{\hat{e}}_i = -k_2 \left(C_i \hat{e}_i - \sum_{j \in \mathcal{N}_i} (\hat{e}_j - \hat{e}_i) \right). \quad (18)$$

Based on the subsequent Lyapunov-based stability analysis, the adaptation law for the Lb-DNN is designed as

$$\dot{\hat{\theta}}_i = \Gamma_i \left(\nabla_{\hat{\theta}_i}^\top \Phi_i(\hat{q}_0^i, \hat{\theta}_i) (C_i^\top \tilde{y}_i + C_i q_i) - k_3 \hat{\theta}_i \right), \quad (19)$$

for all $i \in \mathcal{V}$, where $k_3 \in \mathbb{R}_{>0}$ is a user-defined gain, $\Gamma_i \in \mathbb{R}^{p \times p}$ is a symmetric and positive-definite user-defined gain, and $\nabla_{\hat{\theta}_i} \Phi_i : \mathbb{R}^n \times \mathbb{R}^p \rightarrow \mathbb{R}^{n \times p}$.

V. STABILITY ANALYSIS

The DNN described in (15) is inherently nonlinear with respect to its weights. Addressing NIP uncertainties in such systems, especially when designing adaptive controllers and performing stability analysis, involves significant challenges. To address these complexities, a first-order Taylor approximation for the DNNs is used. To quantify the approximation, the parameter estimation error is defined as

$$\tilde{\theta}_i = \theta_i^* - \hat{\theta}_i. \quad (20)$$

For the DNN in (15), using (20) and applying a first-order Taylor approximation yields

$$\Phi_i(\hat{q}_0^i, \theta_i^*) = \Phi_i(\hat{q}_0^i, \hat{\theta}_i) + \nabla_{\hat{\theta}_i} \Phi_i(\hat{q}_0^i, \hat{\theta}_i) \tilde{\theta}_i + O\left(\|\tilde{\theta}_i\|^2\right). \quad (21)$$

Substituting (15) into (14) and using (21) yields

$$\dot{\hat{q}}_i = \Phi_i(\hat{q}_0^i, \hat{\theta}_i) + \nabla_{\hat{\theta}_i} \Phi_i(\hat{q}_0^i, \hat{\theta}_i) \tilde{\theta}_i + \Delta_i - \dot{\hat{q}}_0^i + \tilde{f}_i, \quad (22)$$

where $\Delta_i \in \mathbb{R}^n$ is defined as $\Delta_i \triangleq O\left(\|\tilde{\theta}_i\|^2\right) + \epsilon_i(\hat{q}_0^i)$.

Substituting (17) into (22), adding and subtracting q_0 within the summation, and using (7) and (12) yields

$$\dot{\hat{q}}_i = \nabla_{\hat{\theta}_i} \Phi_i(\hat{q}_0^i, \hat{\theta}_i) \tilde{\theta}_i + \Delta_i - k_1 \left(C_i \tilde{q}_i - \sum_{j \in \mathcal{N}_i} (\tilde{q}_j - \tilde{q}_i) \right) + \tilde{f}_i. \quad (23)$$

To aid in the stability analysis, the interaction matrix $\mathcal{H} \in \mathbb{R}^{nN \times nN}$ is defined as

$$\mathcal{H} \triangleq (\mathcal{L}_G \otimes I_n) + \mathcal{C}, \quad (24)$$

where $\mathcal{C} \triangleq \sum_{i=1}^N \mathbf{e}_i \mathbf{e}_i^\top \otimes C_i \in \mathbb{R}^{nN \times nN}$, and $\mathbf{e}_i$ is i th standard basis in $\mathbb{R}^N$. Based on the definition of $\mathcal{H}$, applying (24) to (23), yields

$$\dot{\tilde{q}} = \nabla_{\hat{\theta}} \Phi(\hat{q}_0, \hat{\theta}) \tilde{\theta} + \Delta - k_1 \mathcal{H} \tilde{q} + \tilde{f}, \quad (25)$$

where $\tilde{q} \triangleq (\tilde{q}_i)_{i \in \mathcal{V}} \in \mathbb{R}^{nN}$, $\hat{q} \triangleq (\hat{q}_0^i)_{i \in \mathcal{V}} \in \mathbb{R}^{nN}$, $\hat{\theta} \triangleq (\hat{\theta}_i)_{i \in \mathcal{V}} \in \mathbb{R}^{pN}$, $\nabla_{\hat{\theta}} \Phi \triangleq \sum_{i=1}^N \mathbf{e}_i \mathbf{e}_i^\top \otimes \nabla_{\hat{\theta}_i} \Phi_i \in \mathbb{R}^{nN \times pN}$, $\tilde{\theta} \triangleq (\tilde{\theta}_i)_{i \in \mathcal{V}} \in \mathbb{R}^{pN}$, $\Delta \triangleq (\Delta_i)_{i \in \mathcal{V}} \in \mathbb{R}^{nN}$, and $\tilde{f} \triangleq (\tilde{f}_i)_{i \in \mathcal{V}} \in \mathbb{R}^{nN}$. Likewise, applying (24) to (18) yields

$$\dot{\hat{e}} = -k_2 \mathcal{H} \hat{e}, \quad (26)$$

where $\hat{e} \triangleq (\hat{e}_i)_{i \in \mathcal{V}} \in \mathbb{R}^{nN}$. Taking the time-derivative of (20) and using (12) and (19) yields

$$\dot{\tilde{\theta}} = -\Gamma \left(\nabla_{\hat{\theta}}^\top \Phi(\hat{q}, \hat{\theta}) \mathcal{C} \tilde{q} - k_3 \tilde{\theta} \right), \quad (27)$$

where $\Gamma \triangleq \sum_{i=1}^N \mathbf{e}_i \mathbf{e}_i^\top \otimes \Gamma_i \in \mathbb{R}^{pN \times pN}$. Define the concatenated state vector $\xi : \mathbb{R}_{\geq 0} \rightarrow \mathbb{R}^{N(2n+p)}$ as $\xi \triangleq$

$\begin{bmatrix} \tilde{q}^\top & \hat{e}^\top & \tilde{\theta}^\top \end{bmatrix}^\top$. Using (25)-(27), the closed-loop error system can be expressed as

$$\dot{\xi} = \begin{bmatrix} \nabla_{\hat{\theta}} \Phi(\hat{q}, \hat{\theta}) \tilde{\theta} + \Delta - k_1 \mathcal{H} \tilde{q} + \tilde{f} \\ -k_2 \mathcal{H} \hat{e} \\ -\Gamma \left(\nabla_{\hat{\theta}}^\top \Phi(\hat{q}, \hat{\theta}) \mathcal{C} \tilde{q} - k_3 \tilde{\theta} \right) \end{bmatrix}. \quad (28)$$

By Theorem 1, the subsequent stability analysis requires ensuring $\hat{q}_0^i(t) \in \Omega$ for all $i \in \mathcal{V}$, for all $t \in \mathbb{R}_{\geq 0}$. This goal is achieved from a stability result which constrains ξ to a compact domain. Define the compact domain $\mathcal{D} \triangleq \{\xi \in \mathbb{R}^{N(2n+p)} : \|\xi\| \leq \chi\}$ in which ξ is supposed to lie. To facilitate the stability analysis, consider the Lyapunov candidate $V : \mathcal{D} \rightarrow \mathbb{R}_{\geq 0}$ defined as

$$V(\xi) \triangleq \frac{1}{2} \xi^\top P \xi, \quad (29)$$

where $P \triangleq \text{diag}(I_{nN}, I_{nN}, \Gamma^{-1}) \in \mathbb{R}^{N(2n+p) \times N(2n+p)}$. By the Rayleigh-Ritz theorem, (29) satisfies the inequality

$$\lambda_1 \|\xi\|^2 \leq V(\xi) \leq \lambda_2 \|\xi\|^2, \quad (30)$$

where $\lambda_1 \triangleq \frac{1}{2} \min \left\{ 1, \frac{1}{\lambda_{\max}\{\Gamma\}} \right\}$ and $\lambda_2 \triangleq \frac{1}{2} \max \left\{ 1, \frac{1}{\lambda_{\min}\{\Gamma\}} \right\}$. To facilitate the subsequent development, the following assumptions are made.

Assumption 3. [23, Assumption 1] There exists a constant $\bar{\theta} \in \mathbb{R}_{>0}$ such that the unknown ideal weights can be bounded as $\max_{i \in \mathcal{V}} \{\|\theta_i^*\|\} \leq \bar{\theta}$.

Based on the definitions of ξ and $\mathcal{D}$, $\|\tilde{\theta}\| \leq \chi$ whenever $\xi \in \mathcal{D}$. Therefore, $\|\hat{\theta}\| \leq \|\tilde{\theta}\| + \|\theta^*\| \leq \chi + N\bar{\theta}$. Given Assumption 3 and the continuous differentiability of Φ , there exists a constant $\bar{\nabla}_{\hat{\theta}} \Phi \in \mathbb{R}_{>0}$ such that $\|\nabla_{\hat{\theta}} \Phi(\hat{q}, \hat{\theta})\|_F \leq \bar{\nabla}_{\hat{\theta}} \Phi$, whenever $\xi \in \mathcal{D}$. As a result, constants $\bar{\nu}, \bar{\Delta} \in \mathbb{R}_{>0}$ exist satisfying $\|\tilde{\theta}^\top \nabla_{\hat{\theta}}^\top \Phi(\hat{q}, \hat{\theta}) (I_{nN} - \mathcal{C})\| \leq \bar{\nu}$ and $\|\Delta\| \leq \bar{\Delta}$, respectively. Based on the subsequent stability analysis, define $\delta \triangleq \frac{k_3(\chi^2 + N\bar{\theta}^2) + \bar{\nu}^2 + \bar{\Delta}^2}{2}$ and $\lambda_3 = \min \{k_1 \lambda_{\min}\{\mathcal{H}\} - 1 - NL, k_2 \lambda_{\min}\{\mathcal{H}\}, k_3\}$. Moreover, the set of stabilizing initial conditions is defined as $\mathcal{S} \triangleq \{\xi \in \mathcal{D} : \|\xi\| \leq \sqrt{\frac{\lambda_1}{\lambda_2} \chi^2 - \frac{\delta}{\lambda_3}}\}$ and the set $\mathcal{B} \triangleq \{\xi \in \mathbb{R}^{N(2n+p)} : \|\xi\| \leq \sqrt{\frac{\lambda_2 \delta}{\lambda_1 \lambda_3}}\}$ is defined as the set all trajectories converge to.

The positive definiteness of $\mathcal{H}$ is necessary for stability of the closed-loop error system in (28). To this end, the concept of trackability is introduced, which gives a necessary and sufficient condition for the positivity of $\mathcal{H}$. The proofs for the following lemmas are provided in the appendix, and the subsequent theorem states the main result of this work.

Definition 1. (Trackability) A target modeled by (4) is said to be trackable by the multi-agent system defined in (5) if, for any $w \in \mathbb{R}^n$, there exists an agent $i \in \mathcal{V}$ whose output matrix C_i satisfies $C_i w \neq \mathbf{0}_{m_i}$.

Lemma 1. The interaction matrix $\mathcal{H}$ corresponding to a connected and undirected communication topology is positive

definite if and only if the target is trackable by the multi-agent system.

Lemma 2. The target in (4) is trackable by the multi-agent system in (5) if and only if $\text{rank} \left(\sum_{i=1}^N C_i^\top C_i \right) = n$.

Theorem 2. Consider the dynamical systems described by (4) and (5). For any initial conditions of the states $\|\xi(t_0)\| \in \mathcal{S}$, the observer given by (17), the controller given by (16), and the adaptation law given by (19) ensure that ξ exponentially converges to $\mathcal{B}$ in the sense that

$$\|\xi(t)\| \leq \sqrt{\frac{\lambda_2}{\lambda_1}} \sqrt{\|\xi(t_0)\|^2 e^{-\frac{\lambda_2}{\lambda_1}(t-t_0)} + \frac{\delta}{\lambda_3} \left(1 - e^{-\frac{\lambda_2}{\lambda_1}(t-t_0)}\right)} \quad (31)$$

$\forall t \in [t_0, \infty)$, provided that the control gain k_1 is selected such that $k_1 > \frac{1+NL}{\lambda_{\min}\{\mathcal{H}\}}$, and the target is trackable by the multi-agent system.

Proof: Substituting (20) and (28) into the time-derivative of (29) and simplifying yields

$$\begin{aligned} \dot{V} = & -k_1 \tilde{q}^\top \mathcal{H} \tilde{q} - k_2 \tilde{e}^\top \mathcal{H} \tilde{e} - k_3 \tilde{\theta}^\top \tilde{\theta} + k_3 \tilde{\theta}^\top \theta^* \\ & + \tilde{\theta}^\top \nabla_{\hat{\theta}}^\top \Phi(\hat{q}, \hat{\theta}) (I_{nN} - \mathcal{C}) \tilde{q} + \tilde{q}^\top \Delta + \tilde{q}^\top \tilde{f}. \end{aligned} \quad (32)$$

Using Assumption 2 yields the inequality $\tilde{q}^\top \tilde{f} \leq NL \|\tilde{q}\|^2$. Employing Assumption 3 and using Young's inequality yields the bounds $k_3 \tilde{\theta}^\top \theta^* \leq \frac{k_3}{2} (\chi^2 + N \bar{\theta}^2)$, when $\xi \in \mathcal{D}$. Applying Young's inequality yields the bounds $\|\tilde{q}^\top \Delta\| \leq \frac{\|\tilde{q}\|^2}{2} + \frac{\bar{\Delta}^2}{2}$ and $\|\tilde{\theta}^\top \nabla_{\hat{\theta}}^\top \Phi(\hat{q}, \hat{\theta}) (I_{nN} - \mathcal{C})\| \|\tilde{q}\| \leq \bar{\nu} \|\tilde{q}\| \leq \frac{\bar{\nu}^2}{2} + \frac{\|\tilde{q}\|^2}{2}$, when $\xi \in \mathcal{D}$. As a result, (32) can be upper bounded by using the definition of δ and simplifying, as

$$\begin{aligned} \dot{V} \leq & -(k_1 \lambda_{\min}\{\mathcal{H}\} - 1 - NL) \|\tilde{q}\|^2 - k_2 \lambda_{\min}\{\mathcal{H}\} \|\tilde{e}\|^2 \\ & - k_3 \|\tilde{\theta}\|^2 + \delta, \end{aligned} \quad (33)$$

when $\xi \in \mathcal{D}$. Applying the gain conditions for k_1 , using the definition of λ_3 and $\|\xi\|^2$, and using (30) yields that (33) can be upper bounded as

$$\dot{V} \leq -\frac{\lambda_3}{\lambda_2} V(\xi) + \delta, \quad (34)$$

when $\xi \in \mathcal{D}$. Solving the differential inequality in (34) yields

$$V(\xi(t)) \leq V(\xi(t_0)) e^{-\frac{\lambda_3}{\lambda_2}(t-t_0)} + \frac{\lambda_2 \delta}{\lambda_3} \left(1 - e^{-\frac{\lambda_3}{\lambda_2}(t-t_0)}\right), \quad (35)$$

when $\xi \in \mathcal{D}$. Applying (30) to (35) and simplifying yields (31), when $\xi \in \mathcal{D}$.

To ensure $\xi(t) \in \mathcal{D}$ for all $t \in \mathbb{R}_{\geq 0}$, (30) can be used to further upper bound the right hand side of (31) as $\|\xi(t)\| \leq \sqrt{\frac{\lambda_2}{\lambda_1}} \sqrt{\|\xi(t_0)\|^2 + \frac{\delta}{\lambda_3}}$, for all $t \in \mathbb{R}_{\geq 0}$. Since $\mathcal{D} \triangleq \{\zeta \in \mathbb{R}^{3nN} : \|\zeta\| \leq \chi\}$, $\xi(t) \in \mathcal{D}$ always holds if $\sqrt{\frac{\lambda_2}{\lambda_1}} \sqrt{\|\xi(t_0)\|^2 + \frac{\delta}{\lambda_3}} \leq \chi$, which is guaranteed if $\|\xi(t_0)\| \leq \sqrt{\frac{\lambda_1}{\lambda_2} \chi^2 - \frac{\delta}{\lambda_3}}$, i.e. $\xi(t_0) \in \mathcal{S}$. To show $\hat{q}_0^i \in \Omega$ for ensuring the universal function approximation holds, consider the set $\Upsilon \subset \Omega$ defined as $\Upsilon \triangleq \{\zeta \in \Omega : \|\zeta\| \leq \bar{q}_0 + \chi\}$, where $\bar{q}_0$ is defined in Assumption 1. Since $\|\xi\| \leq \chi$ implies $\|\tilde{q}\| \leq \chi$, the

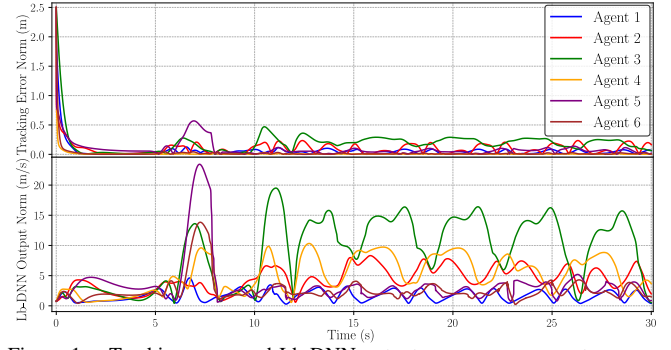


Figure 1. Tracking error and Lb-DNN output response per agent.

following relation holds: $\|\hat{q}_0^i\| \leq \|q_0 - \tilde{q}_i\| \leq \|q_0\| + \|\tilde{q}_i\| \leq \bar{q}_0 + \chi$. Thus, $\hat{q}_0^i(t) \in \Upsilon \subset \Omega$ for all $t \in \mathbb{R}_{\geq 0}$.

Since $\|\xi\| \leq \chi$ implies $\|\tilde{e}\|, \|\tilde{q}\|, \|\tilde{\theta}\| \leq \chi$, $\hat{e}, \hat{q}, \tilde{\theta} \in \mathcal{L}_{\infty}$.

Since $\nabla_{\hat{\theta}} \Phi \in \mathcal{L}_{\infty}$ and Assumption 3 ensures $\hat{\theta} \in \mathcal{L}_{\infty}$, $\hat{\theta} \in \mathcal{L}_{\infty}$. Since $\hat{q}, \hat{\theta} \in \mathcal{L}_{\infty}$, $\Phi \in \mathcal{L}_{\infty}$. Since $q_0, \tilde{q} \in \mathcal{L}_{\infty}$, using (7) yields $\hat{q}_0^i \in \mathcal{L}_{\infty}$, for all $i \in \mathcal{V}$. Similarly, since $\tilde{q} \in \mathcal{L}_{\infty}$, using (12) implies that $\tilde{y}, q_i \in \mathcal{L}_{\infty}$, for all $i \in \mathcal{V}$. Since $\Phi, \hat{q}_0^i, \tilde{y}, q_i \in \mathcal{L}_{\infty}$, using (17) yields $\hat{q}_0^i \in \mathcal{L}_{\infty}$, for all $i \in \mathcal{V}$. Since $\hat{e}, \hat{q}_0^i \in \mathcal{L}_{\infty}$, using (16) implies that $u_i \in \mathcal{L}_{\infty}$. ■

VI. SIMULATION

A simulation is conducted with a network of $N = 6$ agents tasked with tracking a moving target. The agents are initially positioned to form an equilateral N -gon with a radius of 1 meter centered at $(x, y, z) = (0, 0, 0)$. The communication structure is that of a line graph. The target's nonlinear dynamics in (4) are defined by $f(q_0) = [\cos(q_0^x) - \sin(q_0^y) + \cos(2q_0^z); \sin(q_0^y) - q_0^x \cdot q_0^z; q_0^x - q_0^y + q_0^z]$ with initial conditions $(x, y, z) = (0, 0, 0)$, where $q_0 = [q_0^x, q_0^y, q_0^z]$ denotes the spatial coordinates in $\mathbb{R}^3$. The simulation is conducted for a duration of 30 seconds.

Each agent's output matrix $C_i \in \mathbb{R}^{p_i \times 3}$ is generated with entries uniformly distributed between 0.0001 and 1.9999, with $p_i = 2 + (i \bmod 2)$ for $i \in \{1, \dots, 6\}$. Each agent employs an Lb-DNN consisting of 16 hidden layers with 16 neurons per hidden layer, resulting in a total of 4195 weights per agent. The control gains are selected as $k_1 = k_2 = 50$, $k_3 = 0.0001$, and $\Gamma_i = 25 \cdot I_{4195}$ for $i \in \{1, \dots, 6\}$.

Figure 1 details the simulation results, with a mean tracking error of 0.0758. As depicted in Figure 1, all agents effectively track the target with $\|e\| < 0.25$ after approximately 1.0 seconds.

VII. CONCLUSION

In this work, the target tracking problem is addressed for multi-agent systems where the target state information is only partially available to the agents via a heterogeneous measurement model. The concept of trackability is introduced, and an Lb-DNN adaptive controller is developed to achieve target tracking by adaptively compensating for the uncertainty stemming from the unknown target dynamics. A Lyapunov-based stability analysis is provided to guarantee exponential target state estimation and tracking within a neighborhood of the target state.

Proof of Lemma 1: Consider $x \in \mathbb{R}^n$. Since $\mathcal{G}$ is a connected and undirected graph, $(\mathcal{L}_{\mathcal{G}} \otimes I_n)(\mathbf{1}_N \otimes x) = (\mathcal{L}_{\mathcal{G}} \mathbf{1}_N) \otimes (I_n x) = \mathbf{0}_N \otimes I_n x = \mathbf{0}_{nN}$. Therefore, $\ker(\mathcal{L}_{\mathcal{G}} \otimes I_n) = \text{span}\{\mathbf{1}_N \otimes x\}$. By definition, $\mathcal{C} \triangleq \sum_{i=1}^N (\mathbf{e}_i \mathbf{e}_i^\top \otimes C_i) \in \mathbb{R}^{nN \times nN}$ where $\mathbf{e}_i$ is i^{th} standard basis in $\mathbb{R}^N$ and each $C_i \triangleq C_i^\top C_i$ is a Gram matrix and therefore positive semi-definite.

The interaction matrix $\mathcal{H}$ is positive definite if and only if, for any $y \neq \mathbf{0}_{nN}$, $y^\top \mathcal{H} y > 0$. Using the definition of $\mathcal{H}$ and simplifying yields that $\mathcal{H}$ is positive definite if and only if $y^\top (\mathcal{L}_{\mathcal{G}} \otimes I_n) y + y^\top \mathcal{C} y > 0$ which is true if and only if $\ker(\mathcal{L}_{\mathcal{G}} \otimes I_n) \cap \ker(\mathcal{C}) = \emptyset$. Since $\ker(\mathcal{L}_{\mathcal{G}} \otimes I_n) = \text{span}\{\mathbf{1}_N \otimes x\}$, $\ker(\mathcal{L}_{\mathcal{G}} \otimes I_n) \cap \ker(\mathcal{C}) = \emptyset$ if and only if $(\mathbf{1}_N \otimes x) \notin \ker(\mathcal{C})$. It follows that $(\mathbf{1}_N \otimes x) \notin \ker(\mathcal{C})$ if and only if $\mathcal{C}(\mathbf{1}_N \otimes x) \neq \mathbf{0}_{nN}$. Using the definition of $\mathcal{C}$ yields that $\mathcal{C}(\mathbf{1}_N \otimes x) \neq \mathbf{0}_{nN}$ if and only if $(\sum_{i=1}^N (\mathbf{e}_i \mathbf{e}_i^\top \otimes C_i^\top C_i))(\mathbf{1}_N \otimes x) \neq \mathbf{0}_{nN}$ which, by the mixed-product property of the tensor product, is true if and only if $\sum_{i=1}^N ((\mathbf{e}_i \mathbf{e}_i^\top \mathbf{1}_N) \otimes (C_i^\top C_i x)) \neq \mathbf{0}_{nN}$. Using the fact that $\mathbf{e}_i \mathbf{e}_i^\top \mathbf{1}_N = \mathbf{e}_i$ yields that $\sum_{i=1}^N ((\mathbf{e}_i \mathbf{e}_i^\top \mathbf{1}_N) \otimes (C_i^\top C_i x)) = \mathbf{0}_{nN}$ is true if and only if $\sum_{i=1}^N (\mathbf{e}_i \otimes C_i^\top C_i x) \neq \mathbf{0}_{nN}$. Since x is arbitrary, it follows that $\sum_{i=1}^N (\mathbf{e}_i \otimes C_i^\top C_i x) \neq \mathbf{0}_{nN}$ is true, if for any non-zero x , there exists $i \in \{1, 2, \dots, N\}$ such that $x \notin \ker(C_i^\top C_i)$.

To show that $\ker(C_i^\top C_i) = \ker(C_i)$, consider $x \in \ker(C_i^\top C_i)$. By the definition of the kernel, it follows that $x \in \ker(C_i^\top C_i)$ if and only if $C_i^\top C_i x = \mathbf{0}_n$. This condition implies $x^\top C_i^\top C_i x = 0$, and thus $(C_i x)^\top (C_i x) = 0$. Therefore, $C_i x = \mathbf{0}_{m_i}$, implying $x \in \ker(C_i)$. Conversely, if $x \in \ker(C_i)$, then $C_i x = \mathbf{0}_{m_i}$, leading to $C_i^\top C_i x = C_i^\top \mathbf{0}_{m_i} = \mathbf{0}_n$. Hence, x is in $\ker(C_i^\top C_i)$, which establishes $\ker(C_i^\top C_i) = \ker(C_i)$.

Therefore, $\sum_{i=1}^N (\mathbf{e}_i \otimes C_i^\top C_i x) \neq \mathbf{0}_{nN}$ is true, if for any non-zero $x \in \mathbb{R}^n$, there exists $i \in \{1, 2, \dots, N\}$ such that $x \notin \ker(C_i)$. Thus, $\mathcal{H}$ is positive definite if and only if for any non-zero $x \in \mathbb{R}^n$ there exists an $i \in \{1, 2, \dots, N\}$ such that $C_i x \neq \mathbf{0}_{m_i}$. Therefore, based on the definition of trackability, Lemma 1 holds. ■

Proof of Lemma 2: Assume for the sake of contradiction that $\text{rank}(\sum_{i=1}^N C_i^\top C_i) \neq n$. Then, for any non-zero $v \in \mathbb{R}^n$, $v^\top (\sum_{i=1}^N C_i^\top C_i) v = 0$ if and only if $\sum_{i=1}^N v^\top C_i^\top C_i v = 0$ if and only if $\sum_{i=1}^N (C_i v)^\top (C_i v) = 0$. Thus, there exists $i \in \{1, 2, \dots, N\}$ such that $C_i v = \mathbf{0}_{m_i}$, a contradiction. Therefore, trackability implies $\text{rank}(\sum_{i=1}^N C_i^\top C_i) = n$.

Now, suppose $\text{rank}(\sum_{i=1}^N C_i^\top C_i) = n$. Then there exists $i \in \{1, 2, \dots, N\}$ such that $(\sum_{i=1}^N C_i^\top C_i) v \neq \mathbf{0}_n$ which implies that, for some non-zero $v \in \mathbb{R}^n$, $v^\top (\sum_{i=1}^N C_i^\top C_i) v \neq 0$. It follows that $\sum_{i=1}^N (C_i v)^\top (C_i v) \neq 0$ implies that there exists $i \in \{1, 2, \dots, N\}$ such that $C_i v \neq \mathbf{0}_{m_i}$. Therefore, $\text{rank}(\sum_{i=1}^N C_i^\top C_i) = n$ implies trackability. ■

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